CONAN YU

Richmond Hill, Ontario
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Skills

Languages: Python, Java, C++, HTML, JavaScript, CSS
Frameworks & Tools: OpenCV, YOLOv8, Google ADK, Gemini API, FastAPI, Google Pub/Sub, Google Antigravity, SciKit-learn, Pandas, PyTorch, Arduino, Git, GitHub, LaTeX
Focus: Machine learning, agentic AI, embedded/hardware integration

Education

University of Waterloo - Bachelor of Applied Science, Computer Engineering Sept 2026 - June 2031
President's Scholarship of Distinction

Bayview Secondary School - Ontario Secondary School Diploma, IB Diploma Sept 2022 - June 2026
99.8 Top 6 Average · (7/7): IB Math AA SL, Physics SL, English Lit HL, Chemistry HL, Economics HL
(5/5): AP Computer Science A, Calculus BC

Projects

PawSture Aug 2026
Python · OpenCV · YOLOv8-Pose · Arduino · C++
- **1st Place Overall at Waterloo Engineering Design Kickstart**
- Software and hardware embedded real-time posture correction for computer users, using a digital visual/audio indicator and external Arduino OLED
- Detected slouching within **3s with 95% accuracy** across **20+** test users in controlled indoor conditions

Denial Therapy Jun - Aug 2026
Python · Google ADK · Gemini · Pub/Sub · FastAPI · Google Antigravity
- Built an Ambient AI agent (Google ADK, Pub/Sub) that detects health-insurance claim denial events to draft appeal letters for Google's 5-day AI Agents Kaggle Workshop
- Improves estimated appeal letter drafting and turnaround time by **8-10x**
- **100% routing accuracy** on valid datasets using deterministic logic, with prompt injection defense and PII scrubbing

Experience

Aquatic Instructor & Lifeguard, City of Richmond Hill 2024 – 2026
Taught swimming to 100+ students in classes of 1–8, tailoring lesson plans and enforcing facility protocols.

Finance Director, Richmond Hill Youth Action Committee 2023 – 2026
Led finance team to collect $1000+ worth of community sponsorships through email campaigns and in-person outreach. Managed committee event documentation and sponsorship allocation.

Logistics Executive, YR (York Region) Hacks 2023 – 2025
Coordinated event planning and held technical workshops for 500+ participants across two instances of one of Canada's largest hackathons.

Awards & Certifications

Hypatia Waterloo Mathematics Contest: Honour Roll 2025
Canadian Open Mathematics Challenge (COMC): Performance with Distinction 2024
Canadian Intermediate Mathematics Contest (CIMC): Honour Roll 2023

Portfolio

PawSture (team project): github.com/kevinhshen/Five-Cat-PawSture

- Body-tracking ML model detects slouching from a webcam and alert users through both hardware(visual) and software(audio+visual) feedback.
- Program uses EMA-smoothing, auto-calibration of posture, a front/side view detection feature with automatic mode/view swapping, and a browser-based interface hosted on local servers.
- An external Arduino OLED display connected over serial for additional visual alerts when bad posture is sustained.
- 3D printed casing for the Arduino



Figure 1: Arduino OLED showing a neutral posture reading



Figure 2: Front-facing camera mode bad posture



Figure 3: Side-facing camera mode good posture detected on browser interface

Denial Therapy: github.com/ConanY123/Denial-Therapy

- Ambient AI agent that detects health-insurance claim denial events to draft appeal letters, built for Google's 5-day Kaggle AI Agents workshop.
- Event-driven pipeline on Google Pub/Sub to classify incoming denial events, scrubs PII, and includes human review for drafted appeals.
- Keyword-based prompt-injection defenses to protect the LLM creating drafts.

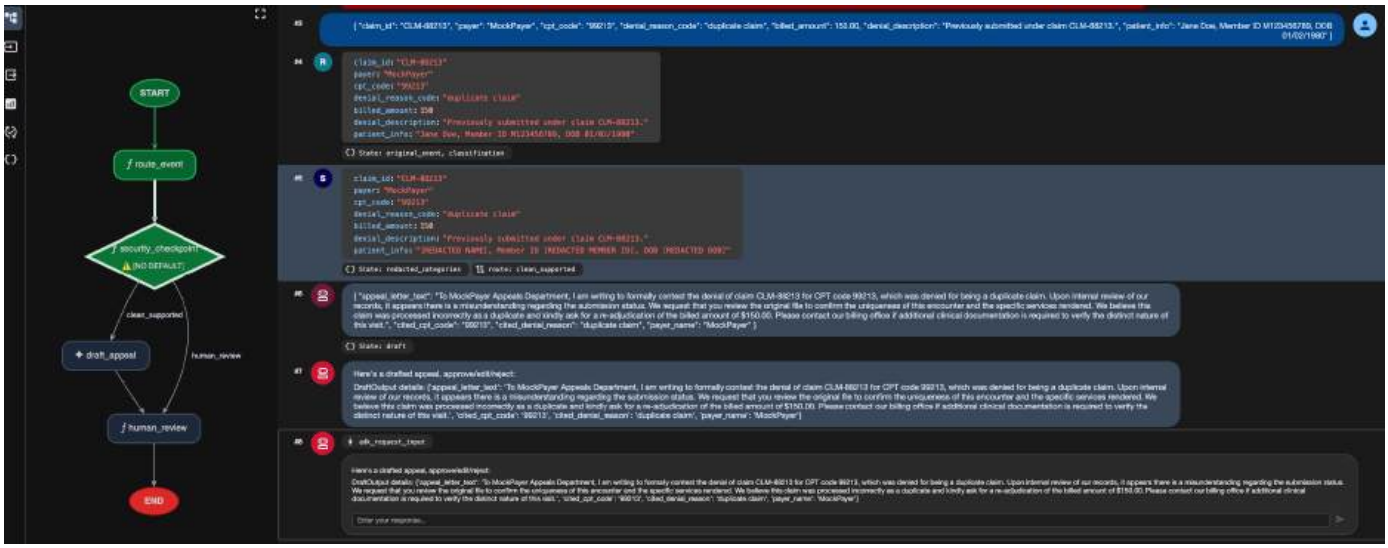


Figure 4: Sample agent trace in ADK with a denial event being classified and turned into a drafted appeal

Portfolio Site: conanyu.ca

- HTML, CSS, JavaScript, Canvas API
- My portfolio site. Content/details centralized in one .js file for page updates without editing markup.
- Hash-based routing for navigation, dark theme with 3 canvas-rendered and swappable/togglable animated rhythm-game backgrounds.





Figures 5 & 6: Homepage with canvas animated and toggleable background